

ARJUN NEMANI

Production Engineer / Site Reliability Engineer

CONTACT

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EDUCATION

International Institute of Information Technology, Hyderabad
B.Tech. in Computer Science with Minor in Computational Linguistics

SKILLS

Reliability
SLI/SLO design, incident response, on-call operations, monitoring, alerting, runbooks, automated remediation

Infrastructure
Linux, Kubernetes, Docker, Terraform, AWS, CI/CD, distributed systems, networking, security

Programming
Python, Go, Bash, Ruby on Rails, JavaScript, SQL, C

Observability
Grafana, metrics pipelines, log analysis, dashboards, load testing, developer tooling

SELECTED PROJECTS

- Encrypted Music Streaming:** integrated proxy re-encryption into decentralized music app.
- HTTP Proxy Server:** multithreaded Python proxy with caching and forwarding.
- Linux Shell:** C shell with piping, redirection, and process control via system calls.
- Video Infra Testing:** load-tested conferencing stack with 100+ simultaneous participants.

KEY ACHIEVEMENTS

- Part of Protocol Labs' APOLLO cohort
- 2nd Place, Gitcoin Protect Privacy Hackathon
- 2nd Place, Gitcoin Spark Hackathon
- CTF Winner, Facebook/Google Bountycon 2019
- Teaching Assistant, Blockchain, IIIT Hyderabad
- Administrator, Open Source Development Group, IIIT-H

PROFILE

Generalist production engineer with broad stack knowledge across reliability, observability, testing, CI/CD, and incident response; strongest in ambiguous production problems that span unfamiliar systems and multiple teams.

WORK EXPERIENCE

Meta, Dublin Apr 2024 - Present
Production Engineer, Host Integrity

- Team builds fleet security infrastructure for TPM-backed host identity, secure erasure, disk encryption, and hardware-backed attestation supporting WhatsApp Private Processing and Incognito Chat with Meta AI.
- Owned operational health for 4 fleet integrity services, defining SLI/SLO standards, alert quality, runbooks, incident readiness, on-call onboarding, and the reliability roadmap.
 - Led fleet-wide remediation with hardware, firmware, OS, and data center partners for 500k+ TPMs affected by vendor-specific shutdown failures, reducing daily failures 96%.
 - Built TPM provisioning workflow with ~25k daily executions, registering TPM certificate identity and validating health/latency before downstream hardware-backed attestation, TLS identity, and disk-encryption workflows.
 - Built low-level TPM tracing and analysis tooling, deep-diving unfamiliar kernel, firmware, hardware, and repair paths to isolate vendor failure modes and guide remediation.
 - Built repair safety net for self-encrypting drives, detecting locked-but-recoverable disks and preventing bad physical repair actions.
 - Caught ~6,000 affected assets and prevented 10,000+ locked drives from unnecessary replacement, protecting capacity during encrypted-drive rollout.
 - Mentored new engineers into production ownership through runbooks, incident reviews, on-call readiness, and hands-on debugging across partner systems.

Meta, Dublin Oct 2022 - Apr 2024
Production Engineer, WhatsApp Production Engineering

- Embedded Production Engineering support model: WhatsApp PEs partnered with product teams; supported WhatsApp Business Flows and the customer-hosted WhatsApp Business On-Premises API.
- Migrated WhatsApp Business on-premises CI from ~30 custom lab machines to on-demand Kubernetes on AWS, reducing diff-time signal from hours to ~10 minutes and lowering infrastructure cost.
 - Built Helm-based Kubernetes CI/CD for the containerized WhatsApp Business client, replacing lab infrastructure and publishing a customer-usable deployment template on GitHub.
 - Supported WhatsApp Business on-premises releases with load testing, release validation, legacy infrastructure decommissioning, and WhatsApp migration planning, reducing on-call toil.
 - Built GA launch-readiness observability for WhatsApp Business Flows across web, API, backend, on-premises, and mobile layers, with 70+ health and performance widgets for on-call and product engineers.
 - Delivered end-to-end latency instrumentation and dashboard optimizations that increased launch confidence, reducing dashboard load time from 105s to 45s.
 - Built interactive runbooks for WhatsApp Flows used 200+ times/year, improving on-call investigations with reusable metric and SQL-driven diagnostics.

Facebook, London Jul 2021 - Sep 2021
Production Engineering Intern

- Built internal documentation-as-code tooling inspired by Sphinx, helping teams maintain docs alongside code in Facebook's monorepo.

Enlume Technologies May 2020 - Jun 2021
Site Reliability Engineer

- Led POC for remote device compliance and logging platform for clients with 25k+ employees and 50k+ devices.
- Scaled Jitsi video conferencing to 2,000+ concurrent participants across 50+ conferences; deployed Terraform-managed video bridges across providers.
- Load-tested video applications with 100+ simultaneous participants and designed hybrid-cloud strategy to optimize infrastructure cost.

Alcrowd Jan 2019 - Apr 2020
Platform Developer & Product Manager

- Maintained Ruby on Rails platform for AI challenge workflows, automated submission evaluation, and user-facing analytics.
- Built product features including insight graphs and user tracking, and managed a team of 4 engineers delivering platform improvements.

IIIT Hyderabad Jun 2018 - Jan 2020
Student Systems Administrator

- Operated Linux, network, and security services for ~4,000 users with 24/7 incident response and remediation.
- Managed institute infrastructure including system administration, network operations, security procedures, and service reliability.